

- 3 (a) moment: force $\times$ perpendicular distance M1
of force from pivot / axis / point A1
couple: (magnitude of) one force $\times$ perpendicular distance M1
between the two forces A1 [4]
(penalise the 'perpendicular' omission once only)
- (b) (i) $W \times 4.8 = (12 \times 84) + (2.5 \times 72)$ C1
 $W = 250 \text{ N}$ (248 N) A1 [2]
- (ii) *either* friction at the pivot *or* small movement of weights B1 [1]
- 4 (a) (i) *either* force $= e \times (V / d)$ *or* $E = V/d$ C1